

MEDLOCATOR-Medicine Availability Locator

PROJECT REPORT

Submitted by

AFNAN T

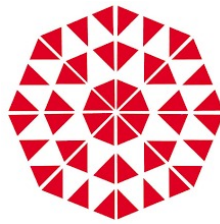
KMC23MCA-2002

to

*the APJ Abdul Kalam Technological University in partial fulfillment of the requirements
for the award of the Degree*

of

Master of Computer Applications



Department Of Management Studies & Computer Applications

KMCT College of Engineering

Kallanthode, NITC P.O, Kozhikode-673601

NOVEMBER 2024

DECLARATION

I undersigned hereby declare that the project report “**MEDLOCATOR-Medicine Availability Locator**”, submitted for partial fulfillment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Mr. Ajayakumar KK**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **MEDLOCATOR-Medicine Availability Locator**”submitted by **AFNAN T (KMC23MCA-2002)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications is a bonafide record of the project work carried out by her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Internal Supervisor

Project Coordinator

Head of the Department

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I would like to take this opportunity to extend my sincere thanks to people who helped me to make this project possible. This project will be incomplete without mentioning all the people who helped me to make it real.

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ABSTRACT

The current digital era, where rapid access to essential healthcare resources is paramount, the role of medical shops and pharmacies has evolved to meet the demands of consumers seeking timely and reliable service. Traditional manual systems in pharmacies often face challenges in efficiency, data management, and quick access to medicine information. A Medicine Availability Locator website aims to address these issues by providing a centralized, automated platform that records and monitors the availability of medicines across various medical shops. This system enables customers to locate specific medications quickly, especially during emergencies or for rare prescriptions, while allowing pharmacies to manage inventory and streamline their operations. Through this initiative, we aim to ensure faster, more reliable access to medications, improving the overall healthcare experience and supporting public health with a user-friendly, accessible solution.

To mitigate these challenges, we propose the development of a Medicine Availability Locator website. This system would allow users to quickly and easily locate specific medications across multiple pharmacies, ensuring timely access to necessary treatments. By enhancing transparency and availability, the platform aims to bridge the gap in medicine accessibility, supporting public health and fostering a more responsive healthcare ecosystem.

Contents

ACKNOWLEDGEMENT	i
ABSTRACT	ii
LIST OF TABLES	vi
LIST OF FIGURES	vii
Chapter 1 INTRODUCTION	1
1.1 General Background	1
1.2 Objective	2
Chapter 2 LITERATURE SURVEY	3
2.1 Literature Survey for Medicine Availability Locator Website	3
Chapter 3 SYSTEM ANALYSIS	6
3.1 Existing System	6
3.2 Proposed System	7
3.3 Module Description	9
3.3.1 Admin	9
3.3.2 Pharmacy	10
3.3.3 User	11
3.4 Feasibility Study	13
3.4.1 Operational Feasibility	13
3.4.2 Technical Feasibility	14
3.4.3 Economic Feasibility	15
3.5 System Environment	16
3.5.1 Developer Requirement	16
3.5.2 User Requirement	16

3.6	Actors and their Roles	17
Chapter 4	SYSTEM DESIGN	20
4.1	Methodology	20
4.2	UML Diagrams	21
4.2.1	Use case	21
4.2.2	Activity Diagram	22
4.2.3	Class Diagram	23
4.3	User Story	24
4.4	Product Backlog	25
4.5	Project plan	26
4.6	Design Phase	27
4.6.1	Login	27
4.6.2	Pharmacy	28
4.6.3	Category	30
4.6.4	Medicine	31
4.6.5	Stock	32
4.7	Form Design	33
4.7.1	Home Page	33
4.7.2	Login Page	34
4.7.3	Pharmacy Registration Page	35
4.7.4	View Medicine	36
4.7.5	Update Medicine details	37
4.8	TESTING AND IMPLEMENTATION	38
4.8.1	Testing	38
4.8.2	Test Case-1 - Admin Login	38
4.8.3	Test Case-2 Admin verify pharmacy details	39
4.8.4	Test Case-3 -Admin manage category	39
4.8.5	Test Case-4 -Pharmacy registration/login	40
4.8.6	Test Case-5 - Pharmacy add / manage medicines	40
4.8.7	Test Case-6 - 6 Pharmacy update stock	41

4.8.8	Test Case-7 - 7 Pharmacy change password	42
4.8.9	Test Case-8 - 8 User search medicines	43
4.8.10	Test Case-9 User check the availability of medicine	43
4.9	Implementation	44
Chapter 5	RESULT AND DISCUSSION	46
Chapter 6	CONCLUSION	48
6.1	Future Work	49
	REFERENCES	50
	APPENDICES	51
6.2	Screenshots	51
6.2.1	Home page for Admin	51
6.2.2	Pharmacy and Admin Login page	52
6.2.3	Admin manage category	53
6.2.4	Admin Verify Pharmacy	54
6.2.5	Pharmacy Home page	55
6.2.6	Pharmacy Profile view and update page	56
6.2.7	Medicine View Page	57
6.2.8	Medicine Details Edit Page	58
6.2.9	Medicine Stock Details Edit Page	59
6.2.10	Add new Medicine Page	60
6.2.11	User Search and Home Page	61

List of Tables

4.1	Test Case-1	38
4.2	Test case-2	39
4.3	Test case-3	39
4.4	Test case-4	40
4.5	Test case-5	40
4.6	Test case-6	41
4.7	Test case-7	42
4.8	Test case-8	43
4.9	Test case-9	43

List of Figures

4.1	Use case	21
4.2	Activity Diagram	22
4.3	Class Diagram	23
4.4	Home Page	33
4.5	Login	34
4.6	Pharmacy Sign Up	35
4.7	View Medicine Page	36
4.8	Update Medicine Page	37
6.1	Admin Home Page	51
6.2	login page	52
6.3	Manage Category	53
6.4	verify pharmacy	54
6.5	pharmacy home page	55
6.6	Profile view	56
6.7	view medicine	57
6.8	edit medicine details	58
6.9	Edit stock details	59
6.10	Add new medicine	60
6.11	user home page	61

Chapter 1

INTRODUCTION

In today's world, medical agencies are trying to provide more reliable and accurate services in their field, offering services to the customers and employees with all the available choices in their interest. Every Medical shops nowadays are trying to computerized its activities to provide better services to its customers. This leads to automate its existing manual system by the help of computerized equipment, and full-fill the requirements, so that their valuable data/information can be stored for a longer period with easy accessing and manipulation. This system enables to manage and record the activities of whole medical shop. However, In today's fast-paced world, access to essential medicines is crucial for maintaining public health and well-being. Moreover, many individuals often face challenges in locating specific medications, especially during emergencies or when dealing with rare prescriptions. The availability of medicines can vary significantly from one pharmacy to another, leading to delays in treatment and potential health risks. To address this issue, we propose the development of a Medicine Availability Locator website.

1.1 General Background

The healthcare sector has seen rapid advancements in technology, leading to improved patient care and more efficient service delivery. This system organized their daily activities like billing, tablets information, stock details and more. However, one persistent problem

remains: the difficulty in quickly locating medicines that are in stock at nearby pharmacies. Traditionally, patients or caregivers must manually contact multiple pharmacies or visit them in person to find the required medication. This process can be time-consuming, frustrating, and, in some cases, life-threatening when urgent treatment is needed. With the increasing penetration of the internet and smartphones, there is a growing demand for digital solutions that can simplify healthcare processes. A Medicine Availability Locator website aims to leverage this digital shift by providing a platform where users can search for specific medicines and instantly find the nearest pharmacy that has them in stock. The purpose of this project is to reduce time consumption and human effort.

1.2 Objective

The primary objective of this mini project is to design and develop a user-friendly website that serves as a reliable tool for locating medicines in real-time. The website will enable users to:

1. Search for Medications: Allow users to search for specific medicines by name or category.
2. Locate Nearby Pharmacies: Display a list of nearby pharmacies where the medication is available.
3. Real-Time Stock Updates: Provide up-to-date information on medicine availability, ensuring that users can access accurate and current data.
4. User Accessibility: Ensure the platform is accessible across various devices, including desktops, tablets, and smartphones.

By achieving these objectives, the Medicine Availability Locator website will serve as a valuable resource for patients, caregivers, and healthcare professionals, streamlining the process of obtaining necessary medications and improving overall healthcare efficiency.

Chapter 2

LITERATURE SURVEY

2.1 Literature Survey for Medicine Availability Locator Website

The availability of essential medicines is a critical component of effective healthcare delivery. Timely access to medications can significantly impact patient outcomes, especially in emergencies or for chronic conditions requiring ongoing treatment. However, patients often face challenges in locating nearby pharmacies that have the required medicines in stock, particularly in areas with limited healthcare resources or during times of high demand

In response to these challenges, digital solutions have emerged that leverage information technology to improve the accessibility and availability of medicines. A medicine availability locator website is one such solution, designed to bridge the gap between patients and pharmacies. This platform aims to provide users with real-time information about the availability of specific medicines at nearby pharmacies, enabling them to make informed decisions quickly. According to Robert E. Hoyt and Ann Yoshihashi [1], Researchers are exploring multiple promising paradigm-shifting ideas. One approach is to recognize the complementary strengths of humans and computers. Humans are good at constructing and processing meaning. In contrast, computers are much better at processing data. Users can leverage this understanding to design systems that harness the data-processing power of computers to present (display) data in ways that make it easier for humans to grasp and

manipulate meaning. This book gives a practical guide covering various aspects of health informatics, including the development and management of IT systems in healthcare.

Furthermore in a literature by World Health Organization (WHO) called White Paper: Enhancing Medicine Availability with Digital Solutions [2], it discusses global strategies for improving medicine availability using digital tools, providing a broad perspective that can inform your project. The importance is notified as follows, The increased use of digital health tools has been particularly notable during the COVID-19 pandemic (Scott et al., 2020), characterised by limited in-person or physical healthcare interactions. The pandemic has further emphasised the urgency of a digital health revolution, with digital health tools acting as potentially powerful means to alleviate pressure on existing health and social care systems (Fagherazzi et al., 2020; Gunasekaran et al., 2021). However, the COVID-19 pandemic has also further emphasised issues of health equity in general (Lyles et al., 2021), and digital health equity in particular (Crawford and Serhal, 2020). The potential of digital health tools notwithstanding, we face key challenges to their rapid evaluation and scalability. Although some tools are effective, their robust evaluation and successful implementation remain limited to specific geographical settings and disease areas (Gordon et al., 2020; Schreiweis et al., 2019). Consequently, digital health tools have yet to significantly and sustainably impact health and wellbeing at a large scale and are yet to be applied equitably across populations and settings with differing needs (Brewer et al., 2020; Rodriguez et al., 2020). To achieve their full potential, digital health tools need to be evaluated using appropriate study designs, integrated into healthcare systems, and effectively scaled. There is a need for a comprehensive platform to openly share evidence-based digital health tools and associated evidence

The Conference Paper: Challenges and Solutions in Implementing Real-Time Medicine Availability Systems [5], This paper presented at an IEEE conference discusses the technical and operational challenges in implementing real-time systems for tracking medicine availability.

The importance of Managing healthcare supply chain is emphasized by Ashwani Kumar,

Venkatesh Mani, Vranda Jain, Himanshu Gupta, and V.G. Venkatesh [3] in their article as The Healthcare supply chain refers to an extensive network of processes and components that ensure timely manufacturing, distribution, and provision of medicines and healthcare supplies to patients Since healthcare is a specialized industry that requires preparedness during unexpected and troubled times like the outbreak of a pandemic, biological/ chemical warfare, etc., its management differs from commercial supply chain management

In a book by Ellen K. Cromley and Sara McLafferty [4] Geographic Information Systems and Public Health: Eliminating Perinatal Disparity. The final two topical chapters (chapters 9 and 10) discuss applications of GIS in the analysis of access to health services and the location of health services. The authors provide a comprehensive and practical introduction to the issues of health-care access, geographical accessibility, the concept of potential access and its operationalization through measurement of distances, activity spaces, methods for using GIS to identify health-facility service areas, and local area variation in health-services utilization. The book discusses the use of GIS in public health, including applications like locating healthcare services and resources, which is relevant to building a medicine locator website.

This literature survey aims to explore existing research, technological frameworks, and case studies related to the development and implementation of a medicine availability locator website. The survey will cover key areas such as healthcare IT systems, inventory management, user interface design, and data security. By examining the current landscape, this survey seeks to identify best practices, challenges, and potential solutions that can inform the development of an efficient and user-friendly platform

Chapter 3

SYSTEM ANALYSIS

3.1 Existing System

A method of purchasing medicine involves visiting a physical pharmacy or drugstore. This has been the primary mode of accessing medications for many years, and it remains common, especially in areas where digital infrastructure is less developed.

Although this approach if followed by more people very commonly it does have some limitations

Limited Accessibility:

Physical pharmacies may have limited hours of operation, making it difficult for patients to access medications outside of these hours, particularly in emergencies.

EXample: A patient needing medication late at night might not be able to access it until the next day if the pharmacy is closed.

Geographical Barriers:

Patients in rural or remote areas may have to travel long distances to reach a pharmacy, which can be a significant inconvenience, especially for those with mobility issues.

Example: In rural areas, patients may need to travel several miles to the nearest pharmacy, which can be challenging during bad weather or if the patient is ill.

Stock Limitations:

Physical pharmacies may not always have the required medication in stock, particularly for less common drugs, leading to delays in treatment.

Example: A patient may have to visit multiple pharmacies to find a specific medication, especially during shortages.

Higher Costs:

Traditional pharmacies may have higher prices compared to online platforms, where customers can easily compare prices and find discounts.

Example: A prescription drug might be more expensive at a local pharmacy compared to what is offered through an online pharmacy with available coupons or bulk discounts.

Time-Consuming:

Visiting a pharmacy in person can be time-consuming, especially if there are long lines, the pharmacy is far away, or the medication needs to be ordered and picked up later.

Example: A patient might spend a significant portion of their day traveling to and waiting at the pharmacy, which could be avoided with online ordering.

Limited Anonymity:

Some patients may feel uncomfortable discussing their medications in a public setting, especially if they are purchasing treatments for sensitive conditions.

Example: A patient buying medication for a condition like STDs might prefer the privacy of an online purchase.

3.2 Proposed System

The proposed system is designed to enhance the accessibility of medicines by connecting users with nearby pharmacies through a dedicated web platform. This system aims to streamline the process of finding and purchasing medications, particularly for those who

need real-time information on medicine availability. The platform will cater to three main user roles: Admin, Pharmacies, and Users, each with specific functionalities and permissions. Below is a detailed description of the proposed system's modules and features.

The development of a medicine availability locator website involves the integration of various technological components and modules. These include an admin module for managing the system, a pharmacy module for updating and maintaining medicine inventory, and a user module for searching and locating medicines. Each module plays a crucial role in ensuring the overall functionality and reliability of the platform.

The proposed Medicine Availability Locator Website will provide an efficient and user-friendly platform that connects users with nearby pharmacies, ensuring timely access to essential medications. By leveraging modern web technologies, the platform will offer real-time data, secure access, and a seamless experience for all stakeholders involved. This project aims to address the common challenges of medicine availability, making healthcare more accessible and convenient for everyone.

3.3 Module Description

The Medicine Availability Locator Website is designed to bridge the gap between users, pharmacies, and administrators by providing an efficient platform for locating and managing the availability of medicines in nearby pharmacies. The system is divided into three main modules: Admin, Pharmacies, and Users. Each module has specific roles and functionalities that contribute to the overall operation of the platform.

3.3.1 Admin

The Admin module is responsible for overseeing and managing the entire system. Admins ensure that pharmacies are properly verified, manage categories of medicines, and monitor user interactions, including feedback and ratings.

Features:

- Login:

- o Secure access to the admin dashboard using authenticated credentials.

- Verify Pharmacy Details:

- o Admins review and verify the details provided by pharmacies during registration.

- o Verification includes checking the validity of pharmacy licenses, contact information, and physical address.

- o Approved pharmacies are granted access to the platform, while unverified or rejected pharmacies are notified

- Add / Manage Category:

- o Ability to create and manage categories of medicines (e.g., Antibiotics, Painkillers, Vitamins)

- o Categories help in organizing medicines, making it easier for users to search and locate specific drugs.

- o Admins can add, edit, or delete categories based on the latest medical standards and user feedback.

- View User Feedback & Ratings:

- o Access to a dashboard displaying user feedback on pharmacies and the overall platform.

- o Admins can review, respond to, or escalate feedback as necessary.

- o User ratings for pharmacies are monitored to ensure quality service and adherence to standards.

3.3.2 Pharmacy

The Pharmacy module allows registered pharmacies to manage their profiles, update their inventory, and interact with the system to ensure that users have access to accurate and up-to-date information about medicine availability.

Features:

- Login:

- o Secure login for pharmacy representatives using unique credentials.

- Update Profile:

- o Pharmacies can update their contact information, operating hours, location, and other relevant details.

- o Profile updates are subject to admin verification to ensure accuracy.

- Add / Manage Medicines Based on Category:

- o Pharmacies can add new medicines to their inventory, categorizing them appropriately (e.g., OTC, Prescription drugs).

- o The platform supports bulk uploads for large inventories.

- o Medicine details include name, dosage, price, and manufacturer information.

- Update Stock Details:

- o Real-time updates of stock levels to reflect current inventory.

- Change Password:

- o Pharmacies can change their password to maintain account security.

- o Passwords must meet complexity requirements to prevent unauthorized access.

3.3.3 User

The User module is designed for the end-users who need to locate nearby pharmacies, search for specific medicines, check availability, and provide feedback on their experience.

Features:

- View Nearby Pharmacies:

- o Users can view a list of pharmacies in their vicinity using geolocation services.

- Search Medicines Based on Category:

- o Users can search for medicines by entering the name or selecting from a category.

- o The search results display available pharmacies, prices, and additional details like dosage forms.

- Check the Availability of Medicines:

- o Users can check real-time availability of medicines at various pharmacies.

- Send Feedback:

- o Users can provide feedback on their experience with specific pharmacies or the overall platform.

- o Feedback can include ratings, comments, and suggestions for improvement.

- o The feedback helps improve service quality and user satisfaction

The described modules collectively ensure the smooth functioning of the Medicine Availability Locator Website. Each module is designed to cater to specific user roles (Admins, Pharmacies, and Users), ensuring that the system is user-friendly, secure, and efficient in helping users find the medicines they need.

3.4 Feasibility Study

Feasibility study is a survey of possibility of computerization based on the next management of objectives related to an analysis of the existing facilities within the system. The information collection that occurs during preliminary investigation examines system feasibility and the likelihood that the system will be beneficial to the organization.

This feasibility study analyzes the viability of developing a Medicine Availability Locator Website. The project aims to create a platform that allows users to locate nearby pharmacies, check the availability of specific medicines, and interact with pharmacies and the system through feedback. The study covers various aspects such as technical, operational, economic, and legal feasibility to ensure the project is realistic, cost-effective, and sustainable.

In this system the tests, which were carried out for feasibility study were technical feasibility and economic feasibility.

1..Operational Feasibility.

2.Technical Feasibility.

3.Economic Feasibility

3.4.1 Operational Feasibility

Objective: Evaluate the operational aspects to determine if the project can function effectively.

User Base:

Pharmacies: The platform needs to attract and register a significant number of pharmacies to ensure comprehensive coverage and reliable service.

End Users: The system must be easy to use to attract and retain a large user base looking for nearby pharmacies and available medicines. Operational Requirements: Admin

Operations: Administrators must verify pharmacy details, manage categories, and monitor feedback to maintain the platform's credibility.

Pharmacy Operations: Pharmacies need to regularly update their profiles, medicine categories, and stock information to provide accurate data to users.

User Operations: Users should easily access the platform, search for medicines, check availability, and provide feedback.

Feasibility: With proper planning, training, and an intuitive interface, operational demands can be met effectively. The project is operationally feasible, provided there is sufficient user and pharmacy on boarding.

3.4.2 Technical Feasibility

Objective: Assess whether the necessary technology and expertise are available to build the website.

Technology Stack: Frontend: HTML, CSS, JavaScript, and frameworks like React or Angular for creating a user-friendly interface.

Backend: Node.js, Python (Django/Flask), or PHP for server-side operations. Database: MySQL, PostgreSQL, or MongoDB for storing user, pharmacy, and medicine data.

API Integration: RESTful APIs for communication between the frontend and backend, and possibly third-party APIs for geolocation services.

Hosting: Cloud platforms like AWS, Azure, or Google Cloud for scalable hosting solutions.

Technical Expertise: Frontend Developers: Skilled in designing and implementing user interface

Backend Developers: Experienced in server-side programming and database management.

Database Administrators: To design and maintain the database schema.

DevOps Engineers: For continuous integration/continuous deployment (CI/CD) and managing cloud infrastructure.

Feasibility: The required technology is readily available, and the skillset needed can be found in most development teams. The project is technically feasible with a focus on scalable architecture and user experience.

3.4.3 Economic Feasibility

Objective: Assess the financial aspects to determine if the project is economically viable.

Initial Costs:

Development: Costs include hiring developers, designers, and project managers, as well as investing in necessary software and hardware.

Hosting and Maintenance: Ongoing costs for cloud services, domain registration, and site maintenance.

Marketing: Budget required for promoting the platform to pharmacies and end users.

Revenue Generation: Subscription Model: Pharmacies could pay a subscription fee to be listed and maintain their profile on the platform.

Advertisement: Pharmacies or pharmaceutical companies could pay for ad placements on the site.

Premium Services: Users or pharmacies might pay for enhanced features, such as prioritized listings or detailed analytics.

Break-Even Analysis: Calculate the number of paying pharmacies and users needed to cover initial costs and generate profit. Explore partnerships with local healthcare providers or pharmacy chains to boost revenue.

Feasibility: The project is economically feasible if a sustainable business model is implemented. Initial costs are manageable, and multiple revenue streams can support the platform's growth and profitability.

3.5 System Environment

The system environment for developing a Medicine Availability Locator Website involves various components and configurations necessary to support the website's functionality. This environment encompasses the hardware, software, database, network, and security aspects needed to build and run the system effectively.

3.5.1 Developer Requirement

3.5.1.1 Hardware Requirement

- Processor : Intel Core i3 or above
- Ram : 4 GB or above
- Storage : 500GB Hard disk

3.5.1.2 Software Requirement

- Operating system : Windows 8 or above
- Front end :HTML, CSS, Bootstrap, Javascript,Dart/Xml
- Back end : python(Django)
- Database : Mysql
- IDE : Android Studio/JetBrains Pycharm
- Web browser : Google Chrome/Firefox

3.5.2 User Requirement

- Any smartphone/Computer/ Laptop
- Stable internet Access

3.6 Actors and their Roles

Admin

Role: The Admin oversees the entire system, ensuring smooth operations, data integrity, and security. The Admin is responsible for managing pharmacies, users, and the overall platform settings.

]Responsibilities:

- Login: Secure access to the admin panel using authentication credentials.
- Verify Pharmacy Details: Review and approve the registration details of pharmacies, ensuring legitimacy and accuracy before allowing them to use the platform.
- Add / Manage Category: Create and update categories for medicines (e.g., pain relief, antibiotics, etc.) to ensure proper organization and searchability.
- View User Feedbacks and Ratings: Monitor user feedback and ratings submitted for pharmacies, addressing issues, and ensuring high service standards.
- Manage Users: Oversee user accounts, including the ability to deactivate or ban users if necessary.

Pharmacy

Role: The Pharmacy actor represents the staff or management of the registered pharmacies who interact with the system to manage their inventory and profile.

Responsibilities:

- Register
- Login: Secure access to the pharmacy's account using authentication credentials.
- Update Profile: Maintain and update the pharmacy's profile, including contact information, address, and operating hours.
- Add / Manage Medicines Based on Category: Add new medicines to the inventory, categorizing them correctly, and managing existing entries.
- Update Stock Details: Regularly update the availability and quantity of medicines in stock to ensure accurate information is presented to users.
- Change Password: Update login credentials to maintain account security.

User

Role: The User is the primary consumer of the service, utilizing the platform to locate pharmacies and check the availability of required medicines

Responsibilities:

- View Nearby Pharmacies: Search for and view pharmacies based on their geographic proximity.
- Search Medicines Based on Category: Search for specific medicines within categorized lists, making it easier to find what they need
- Check the Availability of Medicines: Verify the availability of specific medicines at nearby pharmacies before visiting or placing an order.

- Send Feedback: Provide feedback on pharmacy services or the overall user experience, contributing to platform improvement.
- Send complaint and view reply

Chapter 4

SYSTEM DESIGN

4.1 Methodology

This project follows Agile methodology. Agile software development comprises various approaches to software development under which requirements and solutions evolve through the collaborative effort of self organizing and cross-sectional teams and their customers/ end users. It advocates adaptive planning, evolutionary development, early delivery and continuous improvement and it encourage rapid and flexible response to change. it's a process for managing a project that involves constant collaboration and working in iterations. Today, the word Agile can refer to these values and the frameworks for implementing them, including Scrum. One thing that seperates Agile from other approaches to software development is the focus on the people doing then work and how they work together. Solutions evolve through collaboration between self-organizing cross-functional teams utilizing the appropriate practices for their context.

4.2 UML Diagrams

4.2.1 Use case

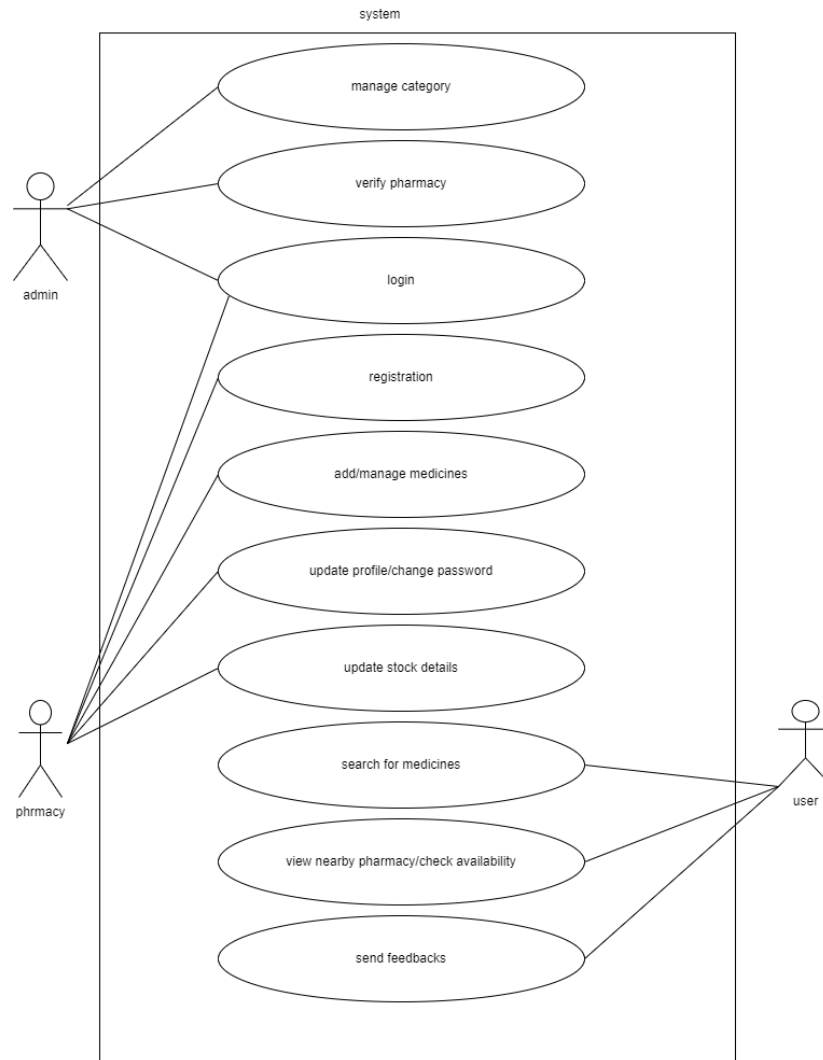


Figure 4.1: Use case

4.2.2 Activity Diagram

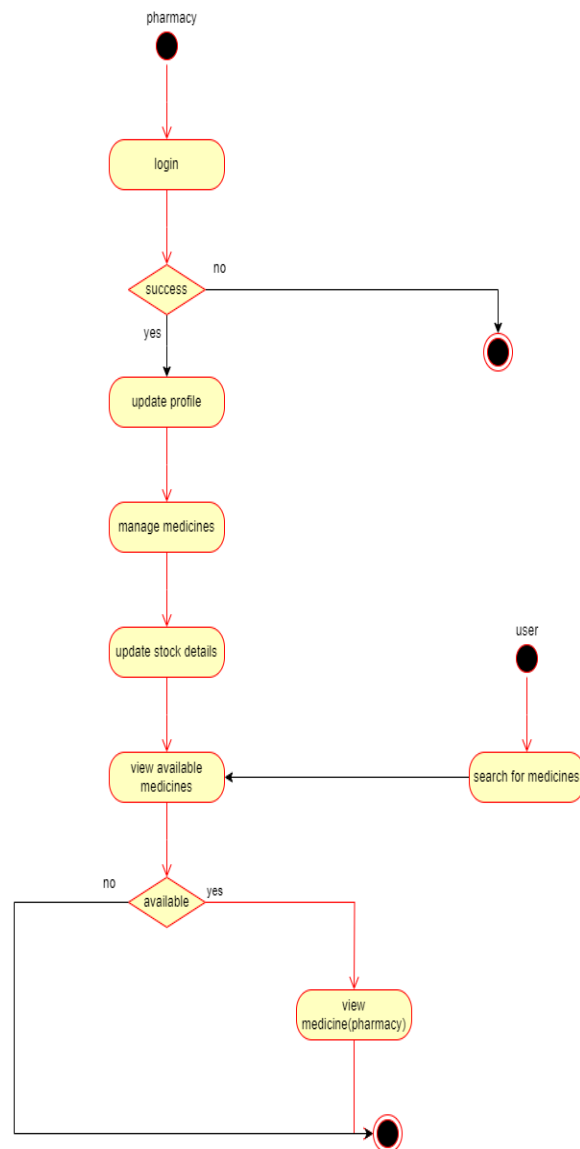


Figure 4.2: Activity Diagram

4.2.3 Class Diagram

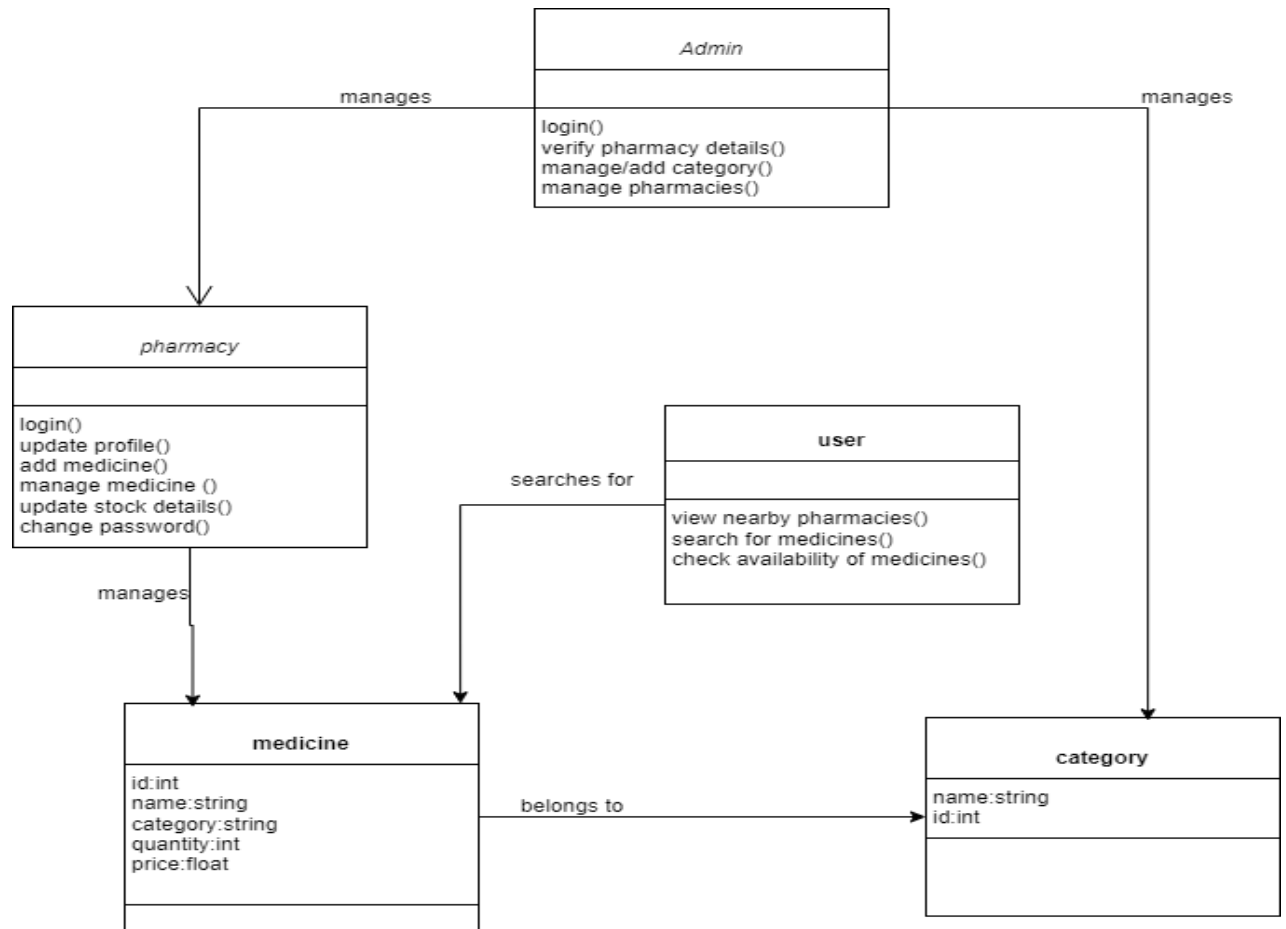


Figure 4.3: Class Diagram

4.3 User Story

User story ID	As a <Type of Users >	I want to <Perform some task >	So that I can <Achieve some goal >
1	Admin	Login/HomePage	To access the system
2	Admin	verify pharmacy details	ensure accuracy of pharmacies
3	Admin	manage category and pharmacies	keep the medicine catalog organized
4	Pharmacy	registration/login/update profile	access and manage pharmacy specific data
5	Pharmacy	add / manage medicines based on category	ensure accuracy and updated medicine details
6	Pharmacy	update stock details	provide real time inventory data to the users
7	Pharmacy	change password	maintain account security
8	User	search medicines based on name/category and view nearby pharmacy	easily find specific type of medicines/find convenient location to purchase medicine
9	User	check the availability of medicine	ensure the medicines are in stock
10	admin	review feedback	can improve the services

4.4 Product Backlog

User Story ID	Priority (Low,High, Medium)	Size	Sprint	Status (Planned, Progressed, Completed)	Release Date	Release Goal
1	MEDIUM	5	1	Planned	20-08-2024	To access the system
2	LOW	2		Planned	02-09-2024	pharmacy details accuray
3	MEDIUM	4		Planned	09-09-2024	manage pharmacy
4	HIGH	8	2	Planned	15-09-2024	pharmacy can login and register
5	MEDIUM	5		Planned	25-09-2024	manages medicine
6	HIGH	9	3	Planned	30-09-2024	live updation
7	LOW	3		Planned	10-10-2024	resett password
8	MEDIUM	6		Planned	21-10-2024	medicines are easily available
9	MEDIUM	7	4	Planned	25-10-2024	can find available medicines
10	LOW	2		Planned	28-10-2024	can review and send feedbacks

4.5 Project plan

User story ID	Task name	Start date	End date	Days	Status Goal
1	Login/homePage	19-08-2024	21-08-2024	3	Planned
2	verify pharmacy details	22-08-2024	25-08-2024	2	Planned
3	manage category and pharmacies	03-09-2024	08-09-2024	5	Planned
4	login/update profile	10-09-2024	20-09-2024	10	Planned
5	add / manage medicines based on category	23-09-2024	24-09-2024	2	Planned
6	update stock details	25-09-2024	30-09-2024	5	Planned
7	change password	01-10-2024	04-10-2024	5	Planned
8	search medicines based on name/category and view nearby pharmacy	10-10-2024	20-10-2024	10	Planned
9	check the availability of medicine	21-10-2024	25-10-2024	4	Planned
10	review feedback	26-10-2024	27-10-2024	2	Planned

4.6 Design Phase

4.6.1 Login

The Login table stores user credentials, including Username, Password, and Type (user role), allowing secure access to the system.

Field Name	Data Type	Description
Username	CharField (max_length=40)	Stores the user's unique identifier or login name.
Password	CharField (max_length=40)	Stores the user's password for authentication.
Type	CharField (max_length=40)	Defines the role or type of user (e.g., Admin, User).

4.6.2 Pharmacy

The Pharmacy table stores information about each pharmacy in the system, including the login credentials (via foreign key to the Login table), name, location (place, post, pincode), contact details (phone and email), and geolocation (latitude and longitude). It also includes the pharmacy's operational status and an image field for uploading the pharmacy's logo or storefront photo. This table helps manage and display pharmacy details for users to find available medicines.

Field Name	Data Type	Description
LOGIN	ForeignKey (Login)	Reference to the Login table, establishes a relationship with login details.
Name	CharField (max_length=40)	Name of the pharmacy, limited to 40 characters.
Place	CharField (max_length=40)	Location or area of the pharmacy, limited to 40 characters.
Post	CharField (max_length=40)	Postal area related to the pharmacy, limited to 40 characters.
status	CharField (max_length=40)	Status of the pharmacy (e.g., active or inactive).
Pincode	IntegerField	Pincode of the pharmacy's location.
Phone_no	BigIntegerField	Contact phone number for the pharmacy.
Email	CharField (max_length=100)	Email address of the pharmacy.
Latitude	CharField (max_length=20)	Latitude coordinate of the pharmacy.
Longitude	CharField (max_length=20)	Longitude coordinate of the pharmacy.
Image	FileField	Field to store an image file associated with the pharmacy.

4.6.3 Category

The Category table is designed to organize and classify items within the system by storing category names and their descriptions. This structure helps streamline data management, making it easier to retrieve and display items based on their respective categories. With fields for the category name and a brief description, this table enhances the application's ability to handle diverse types of data in an organized, accessible format.

Field Name	Data Type	Description
Category	CharField (max_length=30)	Stores the name or title of the category.
Description	CharField (max_length=60)	Provides a brief description of the category.

4.6.4 Medicine

The Medicine table organizes information about available medicines in the system, linking each medicine to a specific pharmacy and category. Key fields include PHARMACY, which references the associated pharmacy; Medicine_name, storing the medicine's name; image, providing a URL or path for the medicine's image; and CATEGORY, linking the medicine to its respective category. This table structure allows for efficient categorization and retrieval of medicines based on pharmacy and category, enhancing the application's usability for locating medicines.

Field Name	Data Type	Description
PHARMACY	ForeignKey (Pharmacy, on_delete=CASCADE)	References the Pharmacy table to associate a medicine with a specific pharmacy.
Medicine_name	CharField (max_length=40)	Stores the name of the medicine.
image	CharField (max_length=400)	Stores the URL or path to the image of the medicine.
CATEGORY	ForeignKey (Category, on_delete=CASCADE)	References the Category table to classify the medicine.

4.6.5 Stock

The Stock table maintains detailed information on medicine inventory, linking each entry to a specific medicine in the system. Key fields include MEDICINE_NAME, which references the medicine associated with the stock entry; Batch_no for tracking batch information; Dosage and expire for dosage and expiration details; description for an extended description of the medicine; Quantity to indicate stock levels; and Price for unit pricing. This structure enables effective tracking of inventory details, ensuring accurate stock management and easy access to critical information about each medicine in storage.

Field Name	Data Type	Description
MEDICINE_NAME	ForeignKey (Medicine, on_delete=CASCADE)	References the Medicine table to associate stock details with a specific medicine.
Batch_no	CharField (max_length=50)	Stores the batch number of the medicine for tracking.
Dosage	CharField (max_length=20)	Specifies the dosage information of the medicine.
expire	CharField (max_length=100)	Indicates the expiration date of the medicine.
description	CharField (max_length=2000)	Provides a detailed description of the medicine.
Quantity	IntegerField	Represents the quantity of the medicine available in stock.
Price	FloatField	Specifies the price per unit of the medicine.

4.7 Form Design

4.7.1 Home Page

The Admin Home page serves as the central dashboard for managing the Medicine Availability Locator website. It provides administrators with an overview of system activities, including real-time statistics on available medicines, pharmacy registrations, and user searches. Key features include options to update the medicine inventory, manage pharmacy accounts, view analytics on user engagement, and handle system notifications. The page is designed with a user-friendly interface, allowing administrators to efficiently monitor and control the platform's functionalities to ensure accurate data and smooth operation.

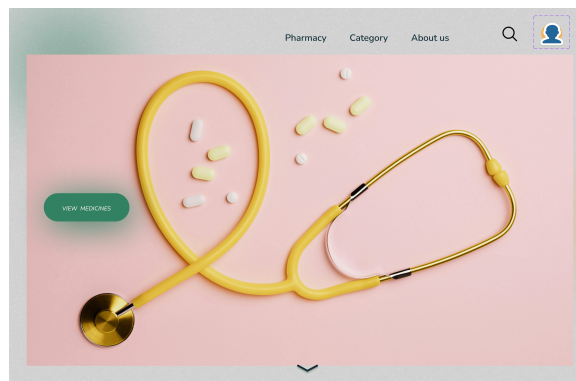


Figure 4.4: Home Page

4.7.2 Login Page

The Login Page is the entry point for users and administrators to access the Medicine Availability Locator website. It requires users to enter their credentials—typically a username or email and password—to authenticate their identity and ensure secure access. The page may include options for "Forgot Password" recovery and links to registration for new users. With a clean and straightforward design, the Login Page ensures a smooth and secure user experience, granting authorized access to the platform's features and personal dashboards.

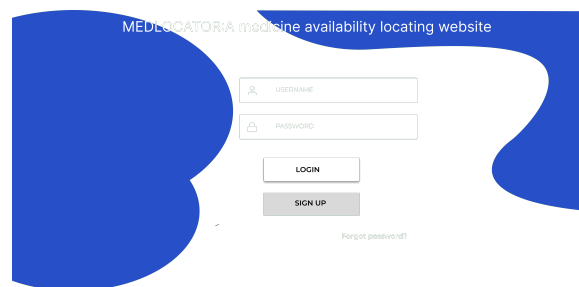
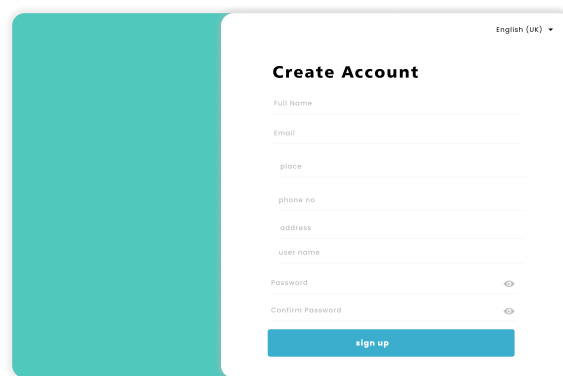


Figure 4.5: Login

4.7.3 Pharmacy Registration Page

The Pharmacy Sign-Up Page is designed for pharmacies to register and join the Medicine Availability Locator platform. This page collects essential information such as the pharmacy's name, location, contact details, operating hours, and administrator credentials. Once registered, the pharmacy can update and manage its inventory in real-time, making its stock visible to users searching for specific medicines. The page is designed with clear instructions and form validation to ensure a seamless and accurate registration process, allowing pharmacies to quickly join and contribute to the platform.



The image shows a 'Create Account' form for pharmacy registration. The form is white with a teal sidebar on the left. It includes fields for Full Name, Email, place, phone no, address, user name, Password, and Confirm Password, followed by a 'sign up' button. A language dropdown shows 'English (UK)'.

Figure 4.6: Pharmacy Sign Up

4.7.4 View Medicine

The View Medicine Page allows users to search for and view detailed information about specific medications available in nearby pharmacies. Users can enter the name of the medicine or browse through categories to check availability, quantity, and pharmacy location. The page may display additional details, such as dosage information, pharmacy contact, and operating hours. Designed for easy navigation, the View Medicine Page ensures users can quickly locate essential medicines and make informed decisions based on real-time availability.

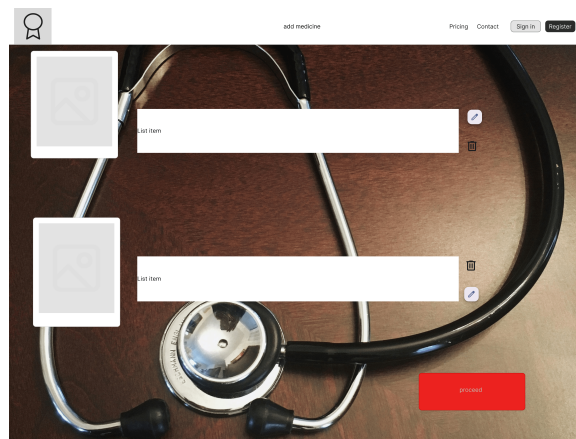


Figure 4.7: View Medicine Page

4.7.5 Update Medicine details

The Update Medicine Page for pharmacies provides an interface for pharmacy staff to edit and maintain details of their existing medicine inventory. On this page, staff can adjust quantities, update expiration dates, change pricing, and add important notes related to specific medicines. The page ensures that any changes made are reflected in real-time, helping to keep inventory information accurate for users searching for medications. This functionality allows pharmacies to manage stock efficiently, ensuring customers see the most up-to-date availability and details.

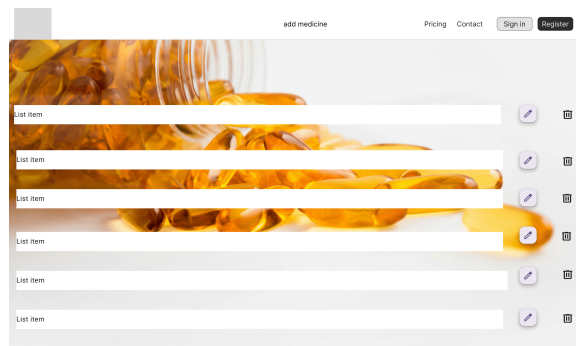


Figure 4.8: Update Medicine Page

4.8 TESTING

AND

IMPLEMENTATION

4.8.1 Testing

4.8.2 Test Case-1 - Admin Login

No.	Date	Action	Expected Result	Actual Result	Pass?
1	28/08/2024	Admin attempts to log in with valid credentials	Admin is redirected to the homepage	Can access the system	Yes
2	29/08/2024	Admin attempts to log in with invalid credentials	Admin receives an error message and remains on login page	Cannot access the system	Yes
3	30/08/2024	Admin leaves the username and password fields blank, then clicks "Login"	Error message prompts admin to enter credentials	cannot login	Yes
4	31/08/2024	Admin logs out from the homepage	Admin is redirected to the login page	logout success	Yes

Table 4.1: Test Case-1

4.8.3 Test Case-2 Admin verify pharmacy details

No.	Date	Action	Expected Result	Actual Result	Pass?
1	03/09/2024	Admin verifies the accuracy of pharmacy details	Pharmacy status is updated to "Verified"	Profile information is successfully verified	Yes
2	03/09/2024	Admin finds incorrect details in pharmacy information and denies verification	Pharmacy is flagged as "Rejected"	Pharmacy is successfully Rejected	Yes

Table 4.2: Test case-2

4.8.4 Test Case-3 -Admin manage category

No.	Date	Action	Expected Result	Actual Result	Pass?
1	05/09/2024	Admin adds a new category	New category is successfully added to the catalog	New category is added successfully.	Yes
2	05/09/2024	Admin edits an existing category name	Category name is successfully updated	Category name is updated successfully.	Yes

Table 4.3: Test case-3

4.8.5 Test**Case-4****-Pharmacy****registration/login**

No.	Date	Action	Expected Result	Actual Result	Pass?
1	10/09/2024	Pharmacy attempts to register with valid details	Pharmacy account is created successfully, and confirmation is shown	Pharmacy is successfully registered with valid details	Yes
2	10/09/2024	Pharmacy logs in with invalid credentials	Error message is displayed, and login is unsuccessful	Pharmacy login unsuccessful	Yes

Table 4.4: Test case-4

4.8.6 Test**Case-5****-****Pharmacy****add****/****manage****medicines**

No.	Date	Action	Expected Result	Actual Result	Pass?
1	17/09/24	Pharmacy adds a new medicine with valid details and category	Medicine is successfully added under the specified category	Medicine is added under the specified category	Yes
2	17/09/24	Pharmacy updates an existing medicine's details	Medicine details are updated successfully	Medicine details are edited	Yes

Table 4.5: Test case-5

4.8.7 Test Case-6 - 6 Pharmacy update stock

No.	Date	Action	Expected Result	Actual Result	Pass?
1	24/09/24	Pharmacy selects a medicine to update stock quantity	Selected medicine details are displayed	Selected medicine details are displayed	Yes
2	24/09/24	Pharmacy enters an invalid stock quantity (e.g., -5) and submits	Error message displayed: "Invalid stock quantity"	stock details are not updated	Yes

Table 4.6: Test case-6

4.8.8 Test Case-7 - 7 Pharmacy change password

No.	Date	Action	Expected Result	Actual Result	Pass?
1	26/09/24	Pharmacy enters incorrect current password	System displays an error indicating the current password is incorrect	'incorrect password' is shown	Yes
2	26/09/24	Pharmacy enters a new password that doesn't meet security requirements (e.g., too short, missing special characters)	System displays an error indicating that the password does not meet security requirements	displays an error indicating that the password does not meet requirements	Yes
3	26/09/24	Pharmacy successfully updates password, logs out, and logs in with new password	Pharmacy can log in with the new password, confirming it was updated correctly	Pharmacy logged in successfully with new password	Yes

Table 4.7: Test case-7

4.8.9 Test Case-8 - 8 User search medicines

No.	Date	Action	Expected Result	Actual Result	Pass?
1	02/10/24	User searches for a medicine by selecting a category	List of medicines within the selected category is displayed	Search results shown successfully	Yes
2	02/10/24	User searches for a non-existent medicine	Message indicating "No results found" is displayed	search result message displayed	Yes

Table 4.8: Test case-8

4.8.10 Test Case-9 User check the availability of medicine

No.	Date	Action	Expected Result	Actual Result	Pass?
1	7/09/24	User checks the availability of a medicine that is in stock	The system confirms the availability of the medicine in stock	medicine in stock is shown	Yes
2	7/10/24	User checks the stock of a medicine in multiple pharmacies	The system displays stock availability in each nearby pharmacy	medicine quantity can be viewed	Yes

Table 4.9: Test case-9

4.9 Implementation

After successful testing, the next phase is Implementation. In this phase, the Medicine Availability Locator system is prepared for deployment, transforming the system design into a fully operational application. Implementation includes setting up the main modules for medicine search, pharmacy inventory management, and user accounts, along with configuring the web application for both users and administrators.

For users, this phase involves finalizing the interface for searching medicines, viewing availability, and locating nearby pharmacies. For administrators, it includes setting up tools for managing pharmacy registrations, updating inventory information, and overseeing user activity on the platform. This phase ensures that all system components work together seamlessly to provide an efficient and user-friendly experience, ready for real-world usage.

Identify Deployment Environment:

To deploy the Medicine Availability Locator system, select a reliable cloud-based platform such as Amazon Web Services (AWS) or Google Cloud Platform (GCP). This ensures scalability, robust security features, and access to reliable database services, which are essential for handling real-time medicine availability data and managing pharmacy inventories. The chosen cloud platform should support automated scaling to manage high traffic during peak times and provide secure data storage to protect user and pharmacy information.

Install Required Software: Set up Node.js for the backend of the web application to handle the core functions, including real-time inventory updates and user account management. Additionally, configure a database, such as MongoDB or PostgreSQL, to store pharmacy details, medicine inventories, and user activity logs

Backend Application Development: Develop the backend using Node.js to handle essential functionalities such as managing medicine inventory data, pharmacy registrations, and user authentication. Implement APIs to facilitate smooth communication between the frontend and backend components.

Web Application Deployment: Create the web application frontend using a modern framework like React for a responsive and user-friendly interface. Integrate the frontend with the Node.js backend for real-time medicine searches, location-based pharmacy lookups, and user account management.

Deployment and Maintenance: Deploy the complete system on the selected cloud platform (AWS/GCP). Provide training materials or sessions for users and administrators to optimize their experience with the platform. Establish a maintenance schedule to perform regular updates, address any bugs, and monitor system performance.

Continuous Improvement: Collect feedback from users and pharmacies to identify potential improvements. Apply updates and new features to meet evolving requirements, ensuring the system remains efficient and user-centric.

Chapter 5

RESULT AND DISCUSSION

The Medicine Availability Locator system automates the process of searching for and locating medicines across multiple pharmacies, providing a seamless and efficient experience for both users and administrators. Through the User Module, customers can easily search for specific medications, check availability in nearby pharmacies, and view real-time stock updates. The system's Admin Module allows administrators to manage pharmacy inventories, update medicine details, and monitor user activity, ensuring accuracy and up-to-date information at all times. Real-time inventory synchronization, search filters by location and medication type, and user-friendly interfaces are key features that contribute to the system's reliability and ease of use.

The Medicine Availability Locator system meets its objectives by improving accessibility to essential medications, reducing the time spent searching for medicines, and enhancing decision-making through accurate, real-time data. Challenges such as varying inventory updates and ensuring data accuracy across pharmacies were addressed through the integration of real-time stock management and system notifications. By empowering users with up-to-date information, the system minimizes delays in treatment and reduces health risks related to unavailable medications.

However, there are some limitations to the current system. While it is effective for urban areas, accessibility in rural or underserved regions could be limited due to internet connectivity or the lack of pharmacy participation. Furthermore, integrating all

pharmacies, especially smaller or independent ones, into the system could present technical challenges. Moving forward, the system can be enhanced by incorporating features like medication price comparisons, delivery tracking, and expanded geolocation services to improve user convenience.

Chapter 6

CONCLUSION

The Medicine Availability Locator system is designed to simplify and improve access to essential medications by bridging the gap between pharmacies and users. Developed with input from healthcare professionals, pharmacies, and end-users, the system addresses the growing need for reliable, real-time information on medicine availability. By enabling users to quickly search for medications, view stock availability, and locate nearby pharmacies, the system enhances accessibility, especially during emergencies or when specialized medications are required. Administrators benefit from automated inventory management and analytics tools, allowing them to efficiently oversee stock levels, pharmacy registrations, and user engagement.

As the Medicine Availability Locator continues to expand, it aims to set a new standard for healthcare accessibility by making medicine information more transparent and accessible. Features such as real-time notifications, personalized user alerts, and automated inventory updates keep users and pharmacies informed, streamlining operations and reducing the risk of stock shortages. Ultimately, the Medicine Availability Locator fosters a more informed, responsive, and accessible healthcare environment, prioritizing timely access to medications, user convenience, and operational efficiency across the pharmaceutical landscape.

6.1 Future Work

Moving forward, the Medicine Availability Locator system aims to incorporate several enhancements to boost functionality, accessibility, and user engagement. One key development will be the launch of mobile applications for both users and administrators, allowing users to search for medications, receive real-time updates, and access pharmacy information directly from their mobile devices. For administrators, these mobile apps will provide remote access to inventory management, allowing them to update stock levels, monitor user activity, and respond to notifications on the go. By extending the system's capabilities to mobile platforms, the Medicine Availability Locator will improve accessibility and make the service more responsive.

In addition, the integration of advanced analytics powered by artificial intelligence (AI) is planned to analyze usage patterns, popular medicine trends, and user feedback. AI-driven insights will help pharmacies manage stock more efficiently, predict high-demand medicines, and optimize inventory to meet user needs. Future enhancements may also include features like medication price comparisons, home delivery tracking, and third-party integrations with healthcare providers to provide a more comprehensive user experience.

Ultimately, these future upgrades are intended to elevate system performance, enhance user convenience, and set a new standard in medicine accessibility by fostering a more adaptable, data-informed, and user-centric healthcare solution.

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APPENDICES

6.2 Screenshots

6.2.1 Home page for Admin

The Admin Home Page in the Medicine Availability Locator system serves as a central dashboard for managing and overseeing key activities across the platform. Here, administrators can view real-time data on medicine availability, pharmacy registrations, and recent user searches. This page includes tools to update inventory, monitor pharmacy participation, and manage user accounts. With quick-access links to inventory management, user analytics, and system notifications, the Admin Home Page ensures that administrators can efficiently maintain accurate, up-to-date information, providing a seamless experience for users and pharmacies alike.

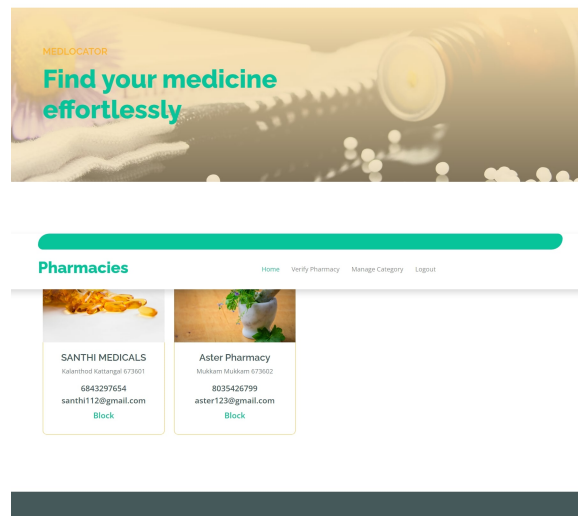


Figure 6.1: Admin Home Page

6.2.2 Pharmacy and Admin Login page

The Pharmacy and Admin Login Page is the secure entry portal for both pharmacy representatives and administrators on the Medicine Availability Locator platform.

Pharmacy staff can log in to update their inventory, manage pharmacy details, and respond to user inquiries, ensuring users see accurate, real-time medicine availability.

Administrators, on the other hand, can log in to access a full range of management tools, including pharmacy monitoring, user account oversight, and system analytics. With a streamlined, user-friendly interface, this login page ensures easy and secure access for both pharmacies and administrators to carry out their respective tasks effectively.

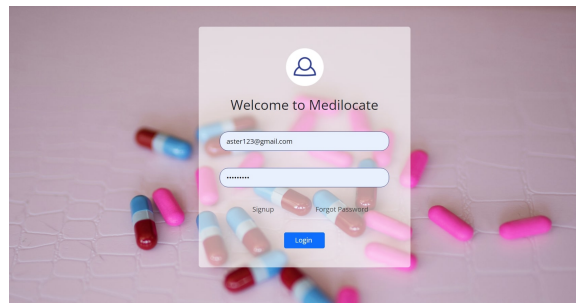


Figure 6.2: login page

6.2.3 Admin manage category

The Admin Manage Category Page in the Medicine Availability Locator system allows administrators to organize and manage medication categories efficiently. This page enables admins to add, edit, or delete categories, ensuring that medicines are organized into clear, searchable groups (e.g., antibiotics, pain relievers, supplements). By maintaining a well-structured category system, this page helps users quickly find the types of medications they need and enhances the overall user experience. The streamlined category management feature also supports better inventory organization and simplifies updates across the platform.

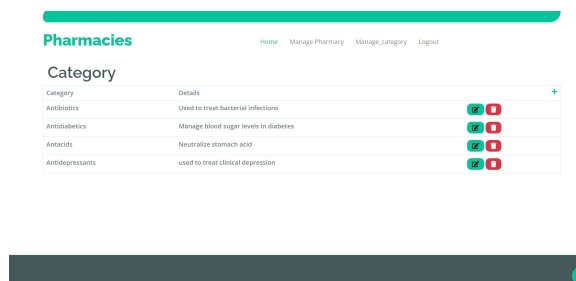


Figure 6.3: Manage Category

6.2.4 Admin Verify Pharmacy

The 'Admin Verify Pharmacy Page' is designed to help administrators review and approve new pharmacy registrations on the Medicine Availability Locator platform. This page displays essential information about each pending pharmacy, such as location, contact details, and credentials, allowing the admin to verify the authenticity and compliance of each pharmacy before granting access. With options to approve, reject, or request further information, this page ensures that only verified and trustworthy pharmacies are added to the platform, maintaining the system's reliability and ensuring safe, accurate information for users.

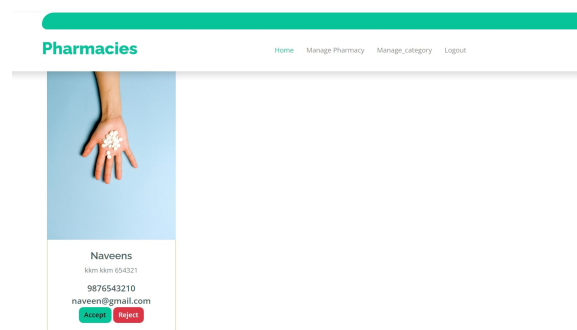


Figure 6.4: verify pharmacy

6.2.5 Pharmacy Home page

The "Pharmacy Home Page" in the Medicine Availability Locator system is a dedicated dashboard for pharmacy staff to manage their medicine inventory and view platform activity. Through this page, pharmacies can update stock levels, add new medicines, and ensure real-time availability data for users. The page also displays recent user interactions, such as searches for medications in their pharmacy, and provides access to notifications and system updates. Designed for easy navigation, the Pharmacy Home Page helps pharmacies keep their information current, improving user access to needed medications and enhancing pharmacy visibility across the platform.

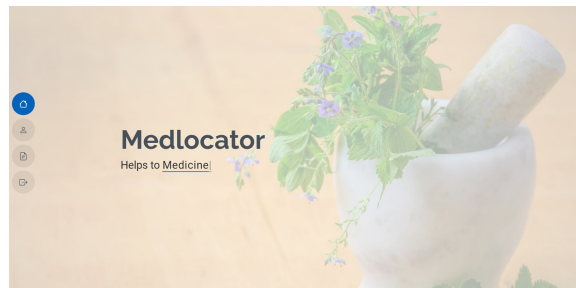
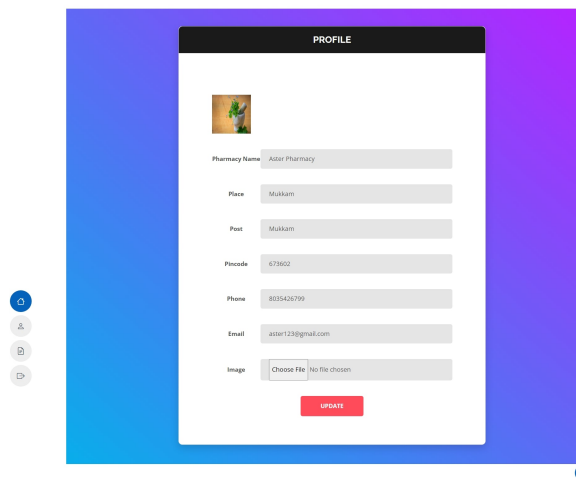


Figure 6.5: pharmacy home page

6.2.6 Pharmacy Profile view and update page

The "Pharmacy Profile View and Update Page" allows administrators and pharmacy owners to view and manage detailed information about each registered pharmacy. This page displays essential details such as the pharmacy's name, location, contact information, and operating hours. Pharmacy owners or administrators can easily update inventory levels, add new medications, or modify existing details to ensure accurate, real-time information is available to users. This streamlined interface helps maintain up-to-date pharmacy profiles, enhancing the reliability of the Medicine Availability Locator system and improving user access to essential medicines.



The screenshot displays a web interface for editing a pharmacy profile. The page has a dark header with the word "PROFILE" in white. Below the header is a profile picture placeholder. The main content area contains a form with the following fields: "Pharmacy Name" (Aster Pharmacy), "Place" (Mukkam), "Post" (Mukkam), "Pincode" (678602), "Phone" (8035426799), "Email" (aster123@gmail.com), and "Image" (Choose file / No file chosen). A red "UPDATE" button is located at the bottom of the form. On the left side of the page, there is a vertical sidebar with four circular icons: a blue one with a plus sign, and three grey ones with a person, a document, and a list icon.

Figure 6.6: Profile view

6.2.7 Medicine View Page

The "Medicine View Page" for pharmacies in the Medicine Availability Locator system provides a clear overview of all medicines in their inventory, organized by 'medicine name' and 'category'. This page allows pharmacy staff to easily browse, update, or manage details for each listed medication, ensuring information such as stock levels and availability remains accurate and current. With a user-friendly layout, this page helps pharmacies maintain organized records and supports efficient inventory updates, enabling them to better serve customers and keep real-time data available on the platform.

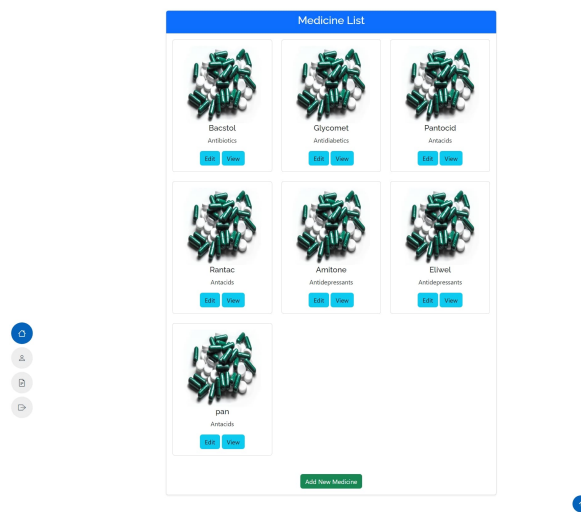


Figure 6.7: view medicine

6.2.8 Medicine Details Edit Page

The "Medicine Details Edit Page" allows pharmacy administrators to update essential information about each medication in their inventory. On this page, admins can modify fields such as the 'medicine name', 'category' (e.g., analgesics, antibiotics), and upload or change the 'medicine image' to ensure accurate identification. This feature enables pharmacies to keep their listings current, ensuring users have access to reliable, detailed information. With an intuitive interface, the Medicine Details Edit Page makes it easy to maintain accurate records, supporting a smooth and informative experience for users searching for medications.

The screenshot shows a web form titled "Edit Medicine Details". It contains several input fields: "Gulliede per" with the value "Rantac", "Batch Number" with "c11", "Dosage" with "150mg", "Quantity" with "30", "Image" with a "Choose File" button and "No file chosen" text, "Expiry Date" with "06/23/2026" and a calendar icon, "Description" with a text area containing "RANTAC 150MG TABLET contains Rantidine which belongs to the group of medicines called H2 receptor antagonists. It is used in the management of", and "Price" with "44". A blue "Save" button is located at the bottom right of the form. A small "Back" button is visible in the top left corner of the page. At the very bottom, there is a link that says "Need Help? Contact Support".

Figure 6.8: edit medicine details

6.2.9 Medicine Stock Details Edit Page

The 'Medicine Stock Details Edit Page' allows pharmacies to manage and update crucial details of each medicine in their inventory. On this page, pharmacy staff can edit information such as dosage, batch number, expiry date, quantity in stock, price, and a description of the medicine. This ensures accurate and up-to-date data for users searching for specific medications. By maintaining these details, pharmacies can provide reliable information on medicine availability, reducing discrepancies and enhancing user trust in the system. The interface is designed to be intuitive, making it easy for staff to quickly update stock as needed.

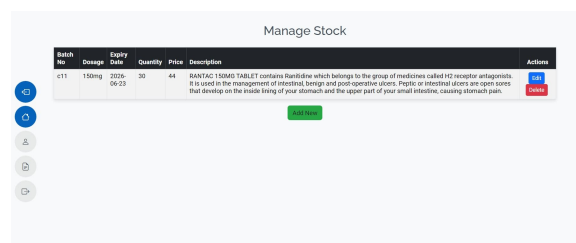


Figure 6.9: Edit stock details

6.2.10 Add new Medicine Page

The "Medicine Details Edit Page" allows pharmacy administrators to update essential information about each medication in their inventory. On this page, admins can modify fields such as the 'medicine name', 'category' (e.g., analgesics, antibiotics), and upload or change the 'medicine image' to ensure accurate identification. This feature enables pharmacies to keep their listings current, ensuring users have access to reliable, detailed information. With an intuitive interface, the Medicine Details Edit Page makes it easy to maintain accurate records, supporting a smooth and informative experience for users searching for medications.

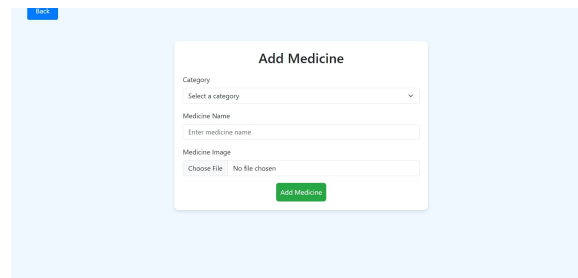
A screenshot of a web application interface for adding a new medicine. The form is titled "Add Medicine" and is centered on a light blue background. It contains four input fields: a dropdown menu for "Category" with the placeholder text "Select a category", a text input for "Medicine Name" with the placeholder text "Enter medicine name", and a file upload area for "Medicine Image" with the placeholder text "Choose File" and "No file chosen". A green "Add Medicine" button is located at the bottom right of the form.

Figure 6.10: Add new medicine

6.2.11 User Search and Home Page

The User Home Page in the Medicine Availability Locator system welcomes users with an intuitive interface to easily access essential features, including a prominent search bar for quick medicine lookups. Users can view nearby pharmacies, check medicine availability. The home page also provides helpful links to account settings and customer support for a user-friendly experience.

The User Search Page allows users to enter the name of a medication to find real-time availability across participating pharmacies. Users can filter results by name or specific medication types, making it easy to find the required medicine quickly. This page enhances accessibility and convenience, ensuring users can locate medications with minimal effort, even during emergencies.

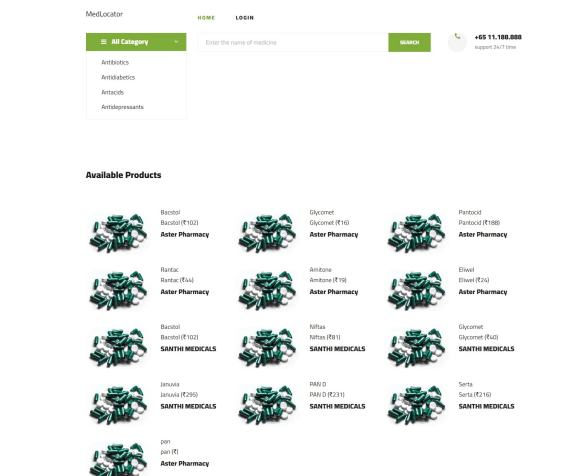


Figure 6.11: user home page